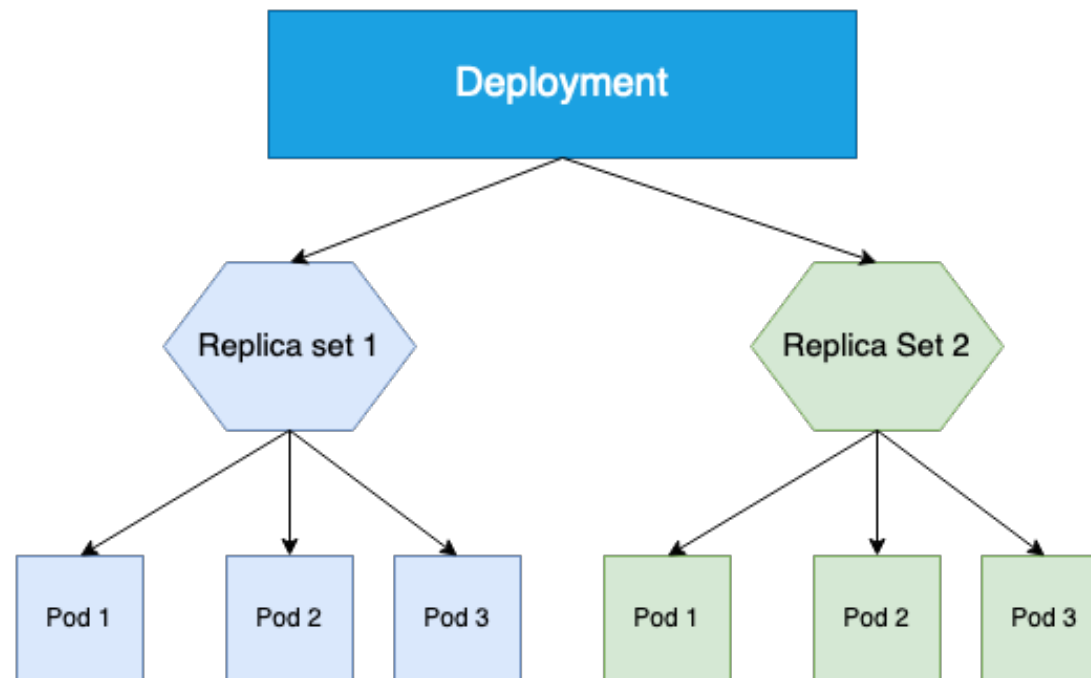


Software Developer
Unità Formativa (UF): Containers, Microservizi -
Serverless
Docente: Denis Maggiorotto
Titolo argomento: Kubernetes Deployments e
Services

Deployment



Deployment

```
deployment-definition.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: myapp-deployment
  labels:
    app: myapp
    type: front-end
spec:
  template:
    metadata:
      name: myapp-pod
      labels:
        app: myapp
        type: front-end
    spec:
      containers:
        - name: nginx-container
          image: nginx
  replicas: 3
  selector:
    matchLabels:
      type: front-end
```

Deployment

```
> kubectl create -f deployment-definition.yml  
deployment "myapp-deployment" created
```

```
> kubectl get deployments
```

NAME	DESIRED	CURRENT	UP-TO-DATE	AVAILABLE	AGE
myapp-deployment	3	3	3	3	21s

```
> kubectl get replicaset
```

NAME	DESIRED	CURRENT	READY	AGE
myapp-deployment-6795844b58	3	3	3	2m

Deployment

```
> kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
myapp-deployment-6795844b58-5rbj1	1/1	Running	0	2m
myapp-deployment-6795844b58-h4w55	1/1	Running	0	2m
myapp-deployment-6795844b58-lfj hv	1/1	Running	0	2m

```
> kubectl get all
```

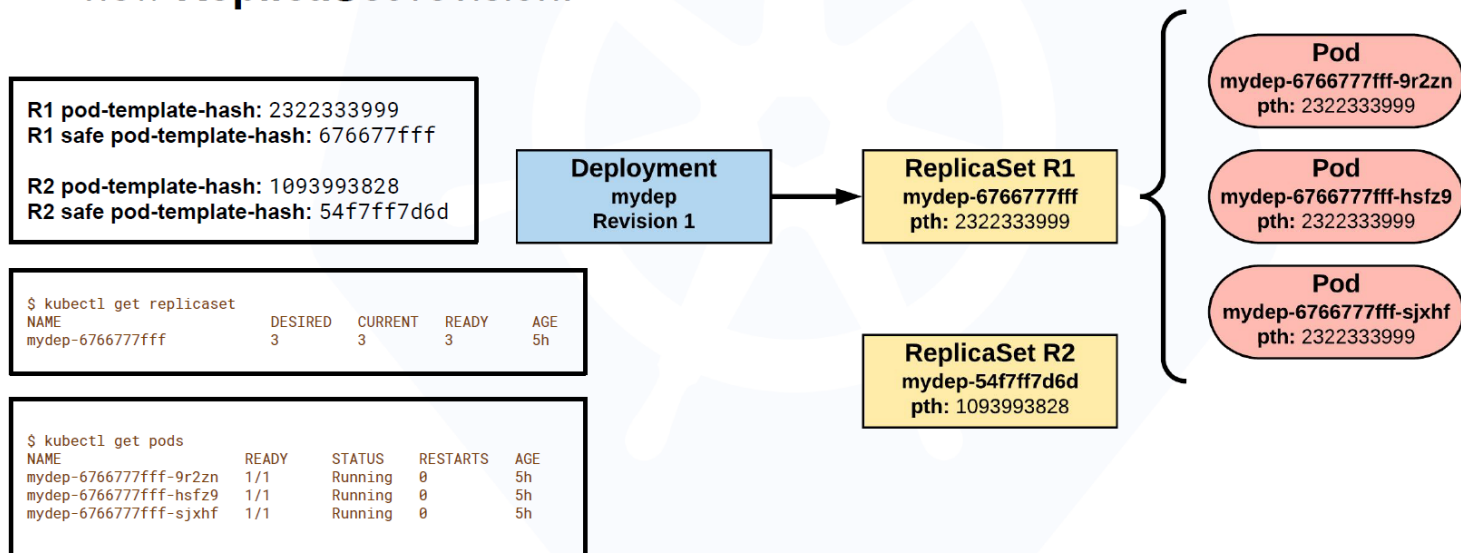
NAME	DESIRED	CURRENT	UP-TO-DATE	AVAILABLE	AGE
deploy/myapp-deployment	3	3	3	3	9h

NAME	DESIRED	CURRENT	READY	AGE
rs/myapp-deployment-6795844b58	3	3	3	9h

NAME	READY	STATUS	RESTARTS	AGE
po/myapp-deployment-6795844b58-5rbj1	1/1	Running	0	9h
po/myapp-deployment-6795844b58-h4w55	1/1	Running	0	9h
po/myapp-deployment-6795844b58-lfj hv	1/1	Running	0	9h

Deployment

Updating pod template generates a new **ReplicaSet** revision.



Deployment

New **ReplicaSet** is initially scaled up based on **maxSurge**.

R1 pod-template-hash: 2322333999
R1 safe pod-template-hash: 6766777fff

R2 pod-template-hash: 1093993828
R2 safe pod-template-hash: 54f7ff7d6d

Deployment
mydep
Revision 2

```
$ kubectl get replicaset
```

NAME	DESIRED	CURRENT	READY	AGE
mydep-54f7ff7d6d	1	1	1	5s
mydep-6766777fff	2	3	3	5h

```
$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
mydep-54f7ff7d6d-9gvll	1/1	Running	0	2s
mydep-6766777fff-9r2zn	1/1	Running	0	5h
mydep-6766777fff-hsfz9	1/1	Running	0	5h
mydep-6766777fff-sjxhf	1/1	Running	0	5h

ReplicaSet R1
mydep-6766777fff
pth: 2322333999

Pod
mydep-6766777fff-9r2zn
pth: 2322333999

Pod
mydep-6766777fff-hsfz9
pth: 2322333999

Pod
mydep-6766777fff-sjxhf
pth: 2322333999

ReplicaSet R2
mydep-54f7ff7d6d
pth: 1093993828

Pod
mydep-54f7ff7d6d-9gvll
pth: 1093993828

Deployment

Phase out of old Pods managed by
`maxSurge` and `maxUnavailable`.

R1 pod-template-hash: 2322333999
R1 safe pod-template-hash: 6766777fff

R2 pod-template-hash: 1093993828
R2 safe pod-template-hash: 54f7ff7d6d

```
$ kubectl get replicaset
NAME                                DESIRED  CURRENT  READY  AGE
mydep-54f7ff7d6d                    2         2        2      8s
mydep-6766777fff                    2         2        2      5h
```

```
$ kubectl get pods
NAME                                READY    STATUS    RESTARTS  AGE
mydep-54f7ff7d6d-9gv1l             1/1      Running   0          5s
mydep-54f7ff7d6d-cqvlq             1/1      Running   0          2s
mydep-6766777fff-9r2zn             1/1      Running   0          5h
mydep-6766777fff-hsfz9             1/1      Running   0          5h
```

Deployment
mydep
Revision 2

ReplicaSet R1
mydep-6766777fff
pth: 2322333999

Pod
mydep-6766777fff-9r2zn
pth: 2322333999

Pod
mydep-6766777fff-hsfz9
pth: 2322333999

~~Pod
mydep-6766777fff-sjxhf
pth: 2322333999~~

ReplicaSet R2
mydep-54f7ff7d6d
pth: 1093993828

Pod
mydep-54f7ff7d6d-9gv1l
pth: 1093993828

Pod
mydep-54f7ff7d6d-cqvlq
pth: 1093993828

Deployment

Phase out of old Pods managed by
maxSurge and **maxUnavailable**.

R1 pod-template-hash: 2322333999
R1 safe pod-template-hash: 6766777fff

R2 pod-template-hash: 1093993828
R2 safe pod-template-hash: 54f7ff7d6d

```
$ kubectl get replicaset
NAME                DESIRED  CURRENT  READY  AGE
mydep-54f7ff7d6d    3        3        3      10s
mydep-6766777fff    0        1        1      5h
```

```
$ kubectl get pods
NAME                READY   STATUS    RESTARTS  AGE
mydep-54f7ff7d6d-9gv1l  1/1     Running   0          7s
mydep-54f7ff7d6d-cqvlq  1/1     Running   0          5s
mydep-54f7ff7d6d-gccr6  1/1     Running   0          2s
mydep-6766777fff-9r2zn  1/1     Running   0          5h
```

Deployment
mydep
Revision 2

ReplicaSet R1
mydep-6766777fff
pth: 2322333999

ReplicaSet R2
mydep-54f7ff7d6d
pth: 1093993828

Pod
mydep-6766777fff-9r2zn
pth: 2322333999

~~Pod
mydep-6766777fff-hsfz9
pth: 2322333999~~

~~Pod
mydep-6766777fff-sjxhf
pth: 2322333999~~

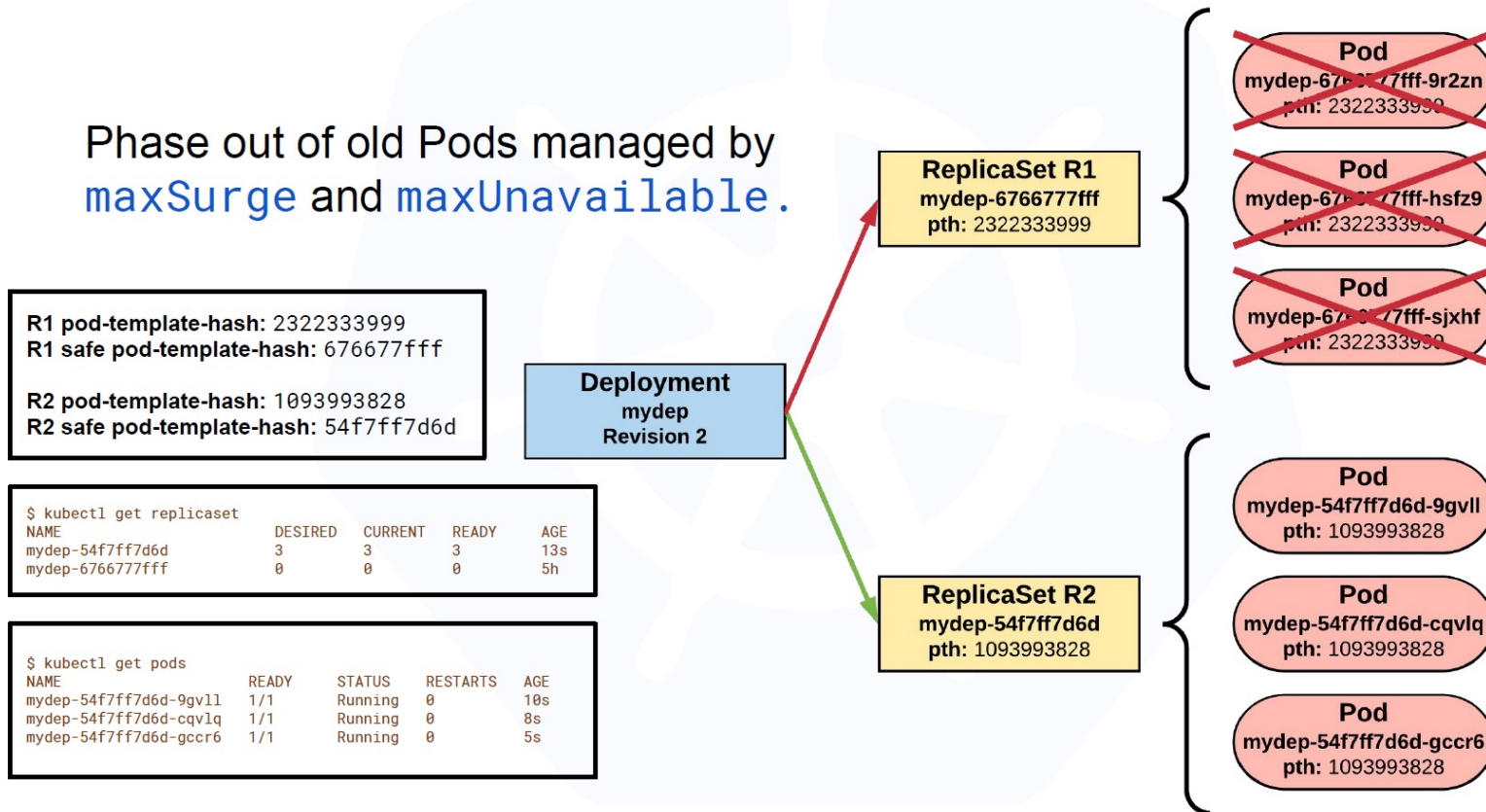
Pod
mydep-54f7ff7d6d-9gv1l
pth: 1093993828

Pod
mydep-54f7ff7d6d-cqvlq
pth: 1093993828

Pod
mydep-54f7ff7d6d-gccr6
pth: 1093993828

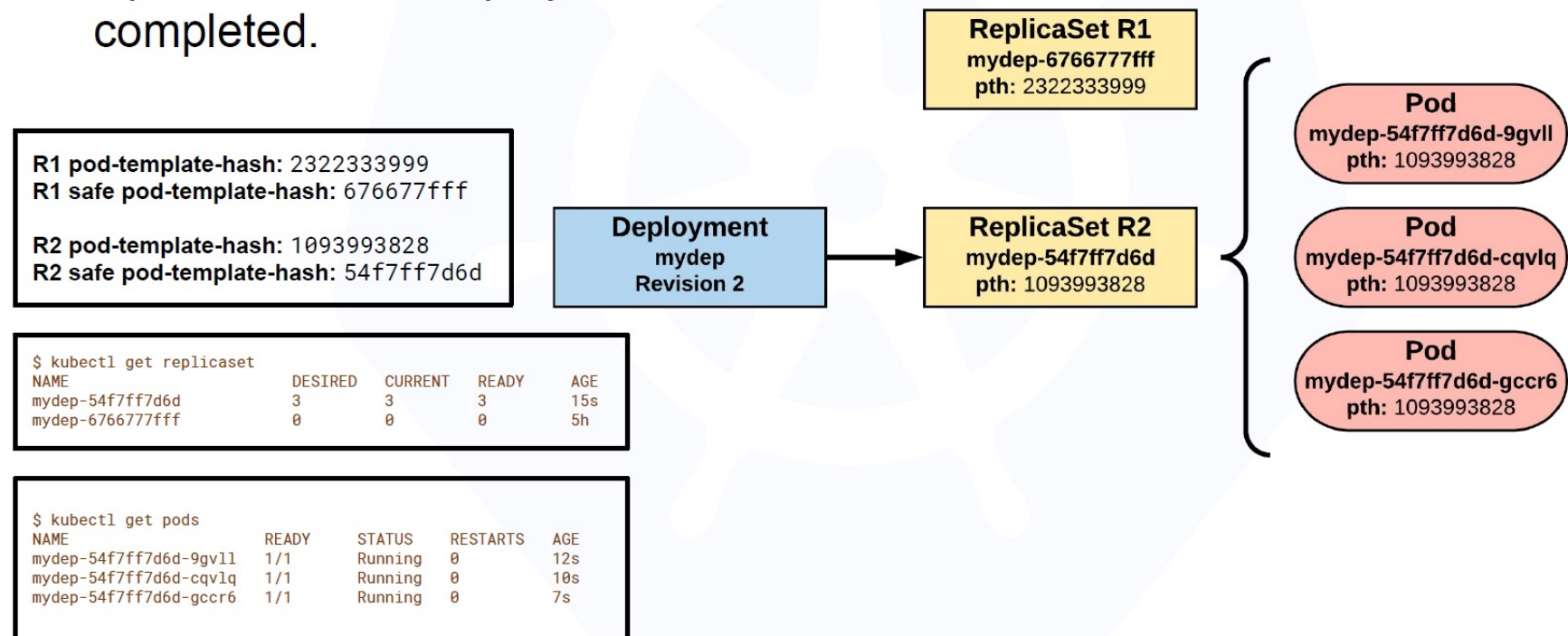
Deployment

Phase out of old Pods managed by
maxSurge and **maxUnavailable**.



Deployment

Updated to new deployment revision completed.

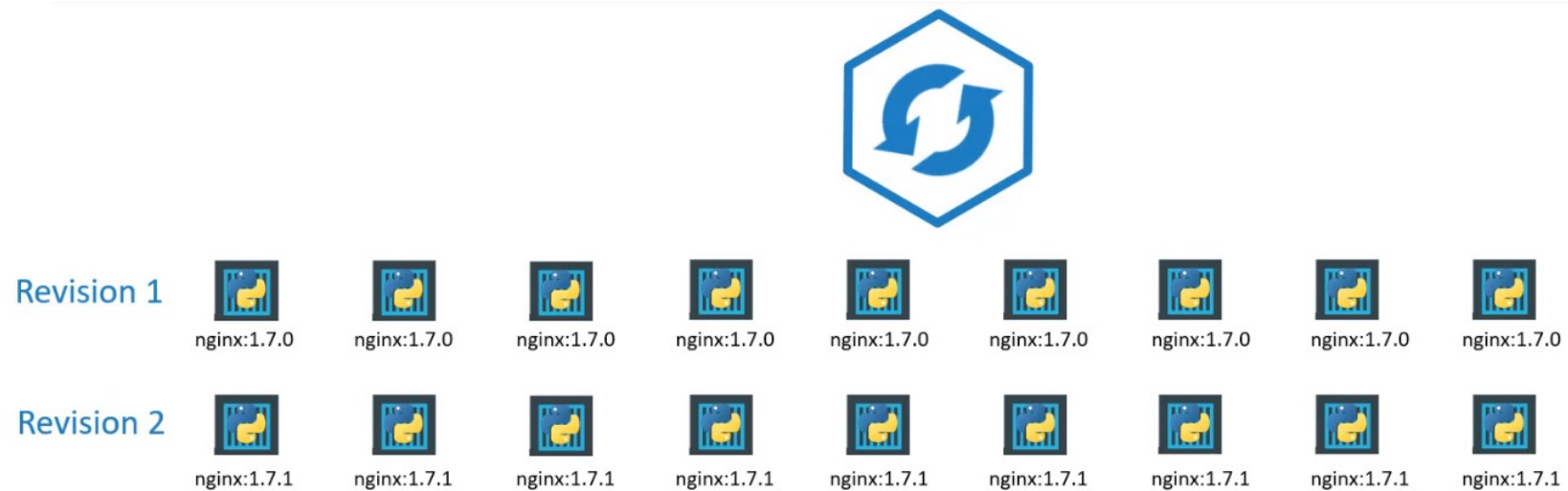


Deployment

maxUnavailable: the maximum number of pods that can be unavailable during the update process. This can be an absolute number or percentage of the replicas count; the default is 25%.

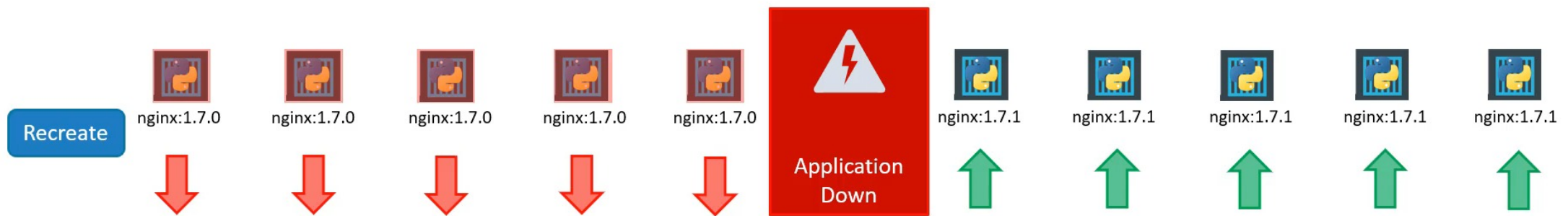
maxSurge: the maximum number of pods that can be created over the desired number of pods. Again this can be an absolute number or a percentage of the replicas count; the default is 25%.

Deployment rollout and versioning



Deployment strategy

Recreate



Deployment strategy

Rolling Update (default)



Deployment strategy

```
#
apiVersion: extensions/v1beta1
kind: Deployment
metadata:
  annotations:
    deployment.kubernetes.io/revision: "1"
  labels:
    run: nginx
  name: nginx
  namespace: default
spec:
  progressDeadlineSeconds: 600
  replicas: 1
  revisionHistoryLimit: 10
  selector:
    matchLabels:
      run: nginx
  strategy:
    type: Recreate
  template:
    metadata:
      creationTimestamp: null
    labels:
      run: nginx
```


Deployment strategy

Recreate

```
Events:
  Type    Reason             Age   From                  Message
  ----    -
Normal    ScalingReplicaSet  11m   deployment-controller Scaled up replica set myapp-deployment-6795844b58 to 5
Normal    ScalingReplicaSet  1m    deployment-controller Scaled down replica set myapp-deployment-6795844b58 to 0
Normal    ScalingReplicaSet  56s   deployment-controller Scaled up replica set myapp-deployment-54c7d6ccc to 5
```

Rolling update

```
Events:
  Type    Reason             Age   From                  Message
  ----    -
Normal    ScalingReplicaSet  1m    deployment-controller Scaled up replica set myapp-deployment-67c749c58c to 5
Normal    ScalingReplicaSet  1s    deployment-controller Scaled up replica set myapp-deployment-7d57dbdb8d to 2
Normal    ScalingReplicaSet  1s    deployment-controller Scaled down replica set myapp-deployment-67c749c58c to 4
Normal    ScalingReplicaSet  1s    deployment-controller Scaled up replica set myapp-deployment-7d57dbdb8d to 3
Normal    ScalingReplicaSet  0s    deployment-controller Scaled down replica set myapp-deployment-67c749c58c to 3
Normal    ScalingReplicaSet  0s    deployment-controller Scaled up replica set myapp-deployment-7d57dbdb8d to 4
Normal    ScalingReplicaSet  0s    deployment-controller Scaled down replica set myapp-deployment-67c749c58c to 2
Normal    ScalingReplicaSet  0s    deployment-controller Scaled up replica set myapp-deployment-7d57dbdb8d to 5
Normal    ScalingReplicaSet  0s    deployment-controller Scaled down replica set myapp-deployment-67c749c58c to 1
```

LAB

https://github.com/sunnyvale-academy/ITS-ICT_Containers

Lab	20 – ReplicationController
Lab	21 – ReplicaSet
Lab	22 – Deployment